But, we note that α is a root of (1). Therefore, we get

$$r\left(-\frac{q}{3r}\right)^3+q\left(-\frac{q}{3r}\right)^2+p\left(-\frac{q}{3r}\right)+1=0\Rightarrow -q^3+3q^3-9pqr+27r^2=0\Rightarrow 9pqr=2q^3+27r^2\,.$$

Example 3.22

It is known that the roots of the equation $x^3-6x^2-4x+24=0$ are in arithmetic progression. Find its roots.

Solution

Let the roots be $a-d,a,a+d$. Then the sum of the roots is $3a$ which is equal to 6 from the given equation. Thus $3a=6$ and hence $a=2$. The product of the roots is a^3-ad^2 which is equal to -24 from the given equation. Substituting the value of a, we get $8-2d^2=-24$ and hence $d=\pm 4$. If we take $d=4$ we get $-2,2,6$ as roots and if we take $d=-4$, we get $6,2,-2$ as roots (same roots given in reverse order) of the equation.

EXERCISE 3.3

1. Solve the cubic equation : $2x^3-x^2-18x+9=0$ if sum of two of its roots vanishes.
2. Solve the equation $9x^3-36x^2+44x-16=0$ if the roots form an arithmetic progression.
3. Solve the equation $3x^3-26x^2+52x-24=0$ if its roots form a geometric progression.
4. Determine k and solve the equation $2x^3-6x^2+3x+k=0$ if one of its roots is twice the sum of the other two roots.
5. Find all zeros of the polynomial $x^6-3x^5-5x^4+22x^3-39x^2-39x+135$, if it is known that $1+2i$ and $\sqrt{3}$ are two of its zeros.
6. Solve the cubic equations : (i) $2x^3-9x^2+10x=3$, (ii) $8x^3-2x^2-7x+3=0$.
7. Solve the equation : $x^4-14x^2+45=0$.

3.7.6 Partly Factored Polynomials

Quartic polynomial equations of the form $(ax+b)(cx+d)(px+q)(rx+s)+k=0$, $k\neq 0$ which can be rewritten in the form $(\alpha x^2+\beta x+\lambda)(\alpha x^2+\beta x+\mu)+k=0$

We illustrate the method of solving this situation in the next two examples.

Example 3.23

Solve the equation

$$(x-2)(x-7)(x-3)(x+2)+19 = 0\,.$$

Solution

We can solve this fourth degree equation by rewriting it suitably and adopting a technique of substitution. Rewriting the equation as

$$(x-2)(x-3)(x-7)(x+2)+19 = 0\,.$$

the given equation becomes

$$(x^2-5x+6)(x^2-5x-14)+19 = 0\,.$$

If we take x^2-5x as y, then the equation becomes $(y+6)(y-14)+19=0$;

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